2. Requirements and user stories

Status: draft derived from the RVL onboarding summary, project test cases and current prototype. Client acceptance is still required.

2.1 Functional requirements

FR-01

Requirement: A user can create a project/object and register sampled sources with identifiers and location/material details.
Acceptance check: Create a fictional project and register at least one sample; the sample is available to the report workflow.
Current evidence/status: Described as implemented in docs/what-we-built.md and app/README.md; verify in live demo because an older status table conflicts.

FR-02

Requirement: A user can register a place that was not sampled with a reason.
Acceptance check: Such a place appears as a survey limitation/PCHM and cannot receive a lab-result link.
Current evidence/status: Implemented according to docs/what-we-built.md; test case TC08.

FR-03

Requirement: A user can enter report lines by paste or upload a PDF.
Acceptance check: Pasted text and a supported digital PDF produce visible lab rows; scan path reports that OCR was used.
Current evidence/status: Implemented for simple layouts; james/extract.py.

FR-04

Requirement: The system extracts report table fields without silently discarding supported columns.
Acceptance check: For a supported table, IDs, descriptions, results and present certificate/method/quantity/remarks columns remain associated with the row.
Current evidence/status: Partial: simple table reconstruction; layout generality and full K1 fields are not established.

FR-05

Requirement: The system proposes a link only to a registered sample.
Acceptance check: An unknown code, duplicate/ambiguous match or conflicting evidence results in no automatic link and an explanation.
Current evidence/status: Implemented in linker/contract/hybrid; synthetic test evidence exists.

FR-06

Requirement: The system uses AI only as assistance for eligible uncertain rows.
Acceptance check: Rules settle clear/safety-conflict cases; invalid model output fails safely; model disagreement does not create an unsafe link.
Current evidence/status: Pipeline implemented against replay client; no live model connected.

FR-07

Requirement: A named inspector can approve, correct, reject or defer each proposal.
Acceptance check: Each recorded decision has a user identity and remains visible in the decision history.
Current evidence/status: Described in app build notes; verify complete flow in demo.

FR-08

Requirement: The system creates a draft IHM report from approved information.
Acceptance check: Approved results appear in relevant draft sections/export; undecided rows do not silently become approved findings.
Current evidence/status: Draft and ODT export described as implemented. Template availability is external to the repo.

FR-09

Requirement: The system flags missing or uncertain information.
Acceptance check: A seeded missing item or unreadable case is shown as a warning or review item, not silently filled.
Current evidence/status: Partial: some completeness checks and page-level OCR note exist; field-level OCR uncertainty is a gap.

FR-10

Requirement: The system preserves evidence about report intake and processing.
Acceptance check: Report source/fingerprint, read method and relevant linker/prompt version can be inspected.
Current evidence/status: Implemented according to build notes.

2.2 Nonfunctional requirements

NFR-01 Safety

Requirement: AI and rules produce proposals only; an authorised person owns final decisions.
Acceptance measure: No report can be treated as final without the required human decision/sign-off.
Current status: Core design principle implemented; confirm roles with RVL.

NFR-02 Safe failure

Requirement: Invalid structured AI output, conflicting evidence or an unregistered sample must not create a link.
Acceptance measure: Deliberately malformed outputs and conflict cases return no link with a reason.
Current status: Covered by contract/linker test design and existing tests; see docs/how-to-test.md.

NFR-03 Traceability

Requirement: A reviewer can reconstruct the input source, processing version and human changes.
Acceptance measure: Evidence and append-only decision history are available for a report.
Current status: Partly implemented; verify retained evidence and access needs with RVL.

NFR-04 Extraction quality

Requirement: Required fields should meet the project target of at least 95% field accuracy.
Acceptance measure: Field-level accuracy measured against expert-labelled representative documents, reported per field and document type.
Current status: Target exists in docs/context.md; current coverage and result do not meet/prove it.

NFR-05 Link quality

Requirement: Target at least 98% correct links and zero silent wrong links on agreed evaluation data.
Acceptance measure: Report correct, wrong, safe-abstain and missed-link counts on a labelled set; separately report hold-out results.
Current status: Synthetic baseline only; not evidence of real-world accuracy.

NFR-06 Usability

Requirement: Inspectors can understand and correct proposals in the workflow.
Acceptance measure: Observe representative users completing tasks; collect task completion, errors and SUS if agreed.
Current status: Not yet measured.

NFR-07 Performance

Requirement: Processing should reduce report handling time by at least 50% versus current practice.
Acceptance measure: Time matched manual and assisted workflows with comparable reports and users.
Current status: Target exists; baseline and outcome not measured.

NFR-08 Privacy

Requirement: No real client or personal data is sent to an external service without explicit approval and suitable controls.
Acceptance measure: Document approved data classes, provider, retention, access and processing terms before any live AI test.
Current status: Fictional data rule is in force; approval not recorded.

NFR-09 Security for hosted pilot

Requirement: Only authorised users can access projects and reports; connections and stored data are protected.
Acceptance measure: Define and verify authentication, role/access policy, HTTPS, secrets management, backup and recovery before pilot use.
Current status: Not demonstrated by the local prototype.

NFR-10 Compatibility

Requirement: Draft output follows the RVL-approved report structure and terminology.
Acceptance measure: RVL reviewer checks a representative generated draft against the approved template.
Current status: ODT export exists; client acceptance and thresholds remain open.

2.3 User stories

1. As an inspector, I want to register a sample with its location, material and source details, so that a lab result can be traced back to where it came from.
2. As an inspector, I want to mark an inaccessible place as not sampled with a reason, so that it is reported as a limitation rather than as confirmed clean.
3. As an inspector, I want to upload or paste a lab report and see the extracted rows, so that I can check the input before matching it.
4. As an inspector, I want a proposed sample link with an understandable reason, so that I can verify the match efficiently.
5. As an inspector, I want unclear or conflicting rows to remain unlinked and be highlighted, so that I can resolve them without a hidden guess.
6. As an authorised reviewer, I want to approve, correct or reject each proposal under my name, so that accountability is clear.
7. As a technical responsible person, I want a draft IHM report assembled from reviewed findings, so that I can check and sign the deliverable.
8. As a quality reviewer, I want to see the report source, processing version and decision history, so that I can trace how the draft was produced.

2.4 Stakeholder questions before baselining

- Which sample-code notation and fields are authoritative?
- Which threshold values and report template version should the prototype use?
- Does each report row carry a certificate number, or is it shared by the report?
- Who decides, who quality-checks and who signs?
- Which report layouts and scan quality are in the first pilot?
- What evidence is sufficient for acceptance of each requirement?

Until answered, these requirements are useful draft acceptance criteria, not a formal client sign-off.